

ArrestShield ML Status

Generated from docs/ML_STATUS.md. Do not edit this PDF directly; edit the markdown and rebuild.

This file is the single source of truth for project status.

docs/ArrestShield_ML_Status_and_Testing_Guide.pdf is generated from it by scripts/build_status_pdf.py. Edit this markdown and rebuild; never edit the PDF directly.

The earlier hand-written PDF (12 August 2026) claimed "ML COMPONENT - Complete" on its front page while its own page 4 listed the reasons production was blocked. Generating the PDF from this file removes that failure mode by construction.

Last verified: 15 August 2026.

One-sentence status

The research pipeline works end to end and now uses a calibrated, seed-stable character model; external scam-call recall is still only 6.17%, so the detector is not promoted and cannot automatically trigger the honeypot.

What is complete

Component	State	Evidence
Dataset pipeline	Complete	52,206 conversations, 615,084 turns, zero split-group leakage
Baseline + ladder models	Complete	SGD and XGBoost, seeds 17/42/93
Calibrated mixed-source candidate	Complete	Character TF-IDF + SGD + Platt scaling; one threshold across three seeds
Frozen external-text evaluation	Complete	243 CC0 scam-call transcripts excluded from training and selection
Multi-task auxiliary heads	Complete	Scam type, stage, 9 tactics
Risk fusion	Complete	XGBoost over out-of-fold base scores
ASR	Complete	Local Whisper-tiny; 0.00 WER on one English clip
Inference API	Complete	/healthz, /v1/model, /v1/detect/text, /v1/detect/audio
LLM honeypot	Complete	Signed handoff, default-deny gate, live Groq engagement verified
Audio -> honeypot chain	Complete	scripts/run_honeypot.py --audio
Tests	98 passing	python -m pytest tests/ -q

Verified end to end on 13 August 2026: audio to Whisper to transcript to trained

detector to HMAC-signed handoff to eligibility gate to live Groq persona reply to a non-evidential transcript.

What is not complete

Limitation	Measurement	Consequence
External scam-call recall	15/243 detected (6.17%); 228 missed	Decisive promotion blocker
External calibration transfer	Brier loss 0.9703 on the external positive-only set	Internally calibrated scores do not transfer to real scammer speech
Financial-dialogue false positives	4.54% on the Banking77 test partition	Passes the configured limit, but Banking77 validation data participated in threshold selection
Hinglish detection quality	Held-out-source ROC-AUC 0.815 +/- 0.023	Improved ranking, still not a deployable accuracy claim
Hinglish evaluation size	21 held-out negatives	Hinglish FPR is not a measurable quantity
Positive label quality	All 2,256 positives source-silver; 71% synthetic	No real-call accuracy claim is possible
Hinglish ASR	Never tested	Only an English clip has been transcribed
ASR speed	38s for 3s of audio on CPU	File upload only; not live monitoring
Human gold set	0 of 150 collected	Final promotion gate unavailable

Corrected headline numbers

Do not quote the 98.74% F1 from `reports/baseline_v1/`. It is a random-split number produced under a corpus shortcut and is not a performance claim.

The defensible candidate numbers are:

- **Held-out-source Hinglish ROC-AUC: 0.815 +/- 0.023** (character features, three seeds, threshold-free)
- **Frozen external scam-call recall: 6.17%** (15/243; positive-only, so FPR cannot be measured)
- **Banking77 test FPR: 4.54%**, with the important caveat that Banking77 validation rows participated in threshold selection
- **Shared threshold: 0.0349148595** across seeds 17/42/93; every eligible negative validation source remains at or below 5% FPR for every seed

The corpus finding

`indian_cyber_scam_phonecall_hinglish` ships 10,000 rows. Of those, 9,257 are exact duplicates. Only 743 unique conversations survive: 633 scam and 110 legitimate.

The Source counts table in docs/datasets/CANONICAL_BUILD_REPORT.md lists input counts, before the 10,140 exact duplicates were removed. Any figure derived from that table overstates the Hinglish corpus by roughly 13x.

The shortcut, and the fixes that worked

49,950 of 52,206 conversations come from three corpora containing only legitimate examples. "Does this look like banking77 / DailyDialog / schema-guided?" was therefore close to a perfect label, and the detector learned it.

Restricting training to sources containing **both** labels removed the shortcut by construction:

Held-out source	Full corpus (36,101 train)	Mixed-only (2,368 train)
indian_cyber_scam_hinglish	0.550 +/- 0.043	0.756 +/- 0.055
synthetic_multi_agent	0.979 +/- 0.009	1.000 +/- 0.000
synthetic_scam_dialogue	0.971 +/- 0.012	1.000 +/- 0.000

Deleting 94% of the training data improved held-out Hinglish AUC by 0.21, roughly four times the seed noise. Reproduce with `python scripts/train_mixed_source_detector.py`.

The follow-up ablation compared word, character, and combined TF-IDF features using source holdouts inside the training partition only. Character features scored 0.880 +/- 0.018 on the training-held-out Hinglish source, versus 0.711 +/- 0.013 for word features and 0.818 +/- 0.029 for combined features. After selection, an untouched test-source audit measured 0.815 +/- 0.023 for the selected character family. Platt calibration reduced internal validation Brier loss from 0.13138 to 0.00428. Calibration and threshold rows are disjoint, and a single threshold is used for all three seeds. Reproduce the complete candidate with `python scripts/train_mixed_source_candidate.py`.

Frozen external evaluation

The CC0 YouTube scam-call corpus is imported into a separate external manifest. Its records are marked ineligible for training, calibration, threshold selection, and model-family selection. Exact transcript duplicates are removed. It contains 243 English positive conversations, so it measures recall only.

Frozen detector	Detected	Missed	Recall
Previous served SGD	3	240	1.23%
Calibrated character candidate	15	228	6.17%

The five-fold relative improvement is useful engineering evidence, but the absolute miss rate is 93.8%. The API therefore reports `research_only_not_promoted`, risk fusion is disabled by policy, and honeypot handoff remains disabled.

Approaches tested and rejected

Both were expected to help and did not. They are recorded because a negative result from a controlled experiment is evidence.

Approach	Result	Why it was dropped
LLM-generated Hinglish augmentation (Groq)	AUC 0.756 -> 0.743	No gain at n=8; a third synthetic style does not add information
External BFSI Hinglish hard negatives (2,452 rows)	AUC 0.756 -> 0.615	Single-label source reintroduced the shortcut it was meant to remove

The second is the more instructive: adding legitimate in-domain conversation *hurt*, because it arrived as a source containing no scam examples. The requirement is legitimate financial conversation from a source that also contains scams - which is what collection has to provide.

Boundary guarantee

No LLM produces a feature, score, threshold, or label anywhere in detection.

- `src/arrestshield/honeypot.py` imports no detector module; a test enforces it.
- The honeypot requires an HMAC-signed detector event and cannot be reached otherwise.
- Honeypot transcripts are stamped `excluded_from_detector_training: true`.
- Whisper is a trained ASR model, not an LLM, and produces words only.
- LLM-generated training rows are provenance: `llm_synthetic`, train-split only, weighted 0.5, and structurally barred from validation and test.

How to verify on this PC

```
.venv\Scripts\python.exe -m pytest tests\ -q
.venv\Scripts\python.exe scripts\train_mixed_source_candidate.py
.venv\Scripts\python.exe scripts\evaluate_external_text.py
.venv\Scripts\python.exe scripts\run_api.py
.venv\Scripts\python.exe scripts\run_honeypot.py --demo --research-mode
```

The honeypot requires `GROQ_API_KEY` in a git-ignored `.env`. Without it every other command still runs.